APLICACIONES
IMÁGENES Y AUDIO

73.64 - TEMAS AVANZADOS EN DEEP LEARNING - 2026

Ing. Eugenia Piñeiro, Ing. Marina Fuster

TABLA DE CONTENIDOS

01. IMÁGENES

02. AUDIO

03. MULTIMODAL

04. EXTRA

05. BIBLIOGRAFÍA

01

IMÁGENES

Aplicaciones

VISION TRANSFORMER (ViT)

[An Image is Worth 16x16 Words]

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]

[*]equal technical contribution, [†]equal advising

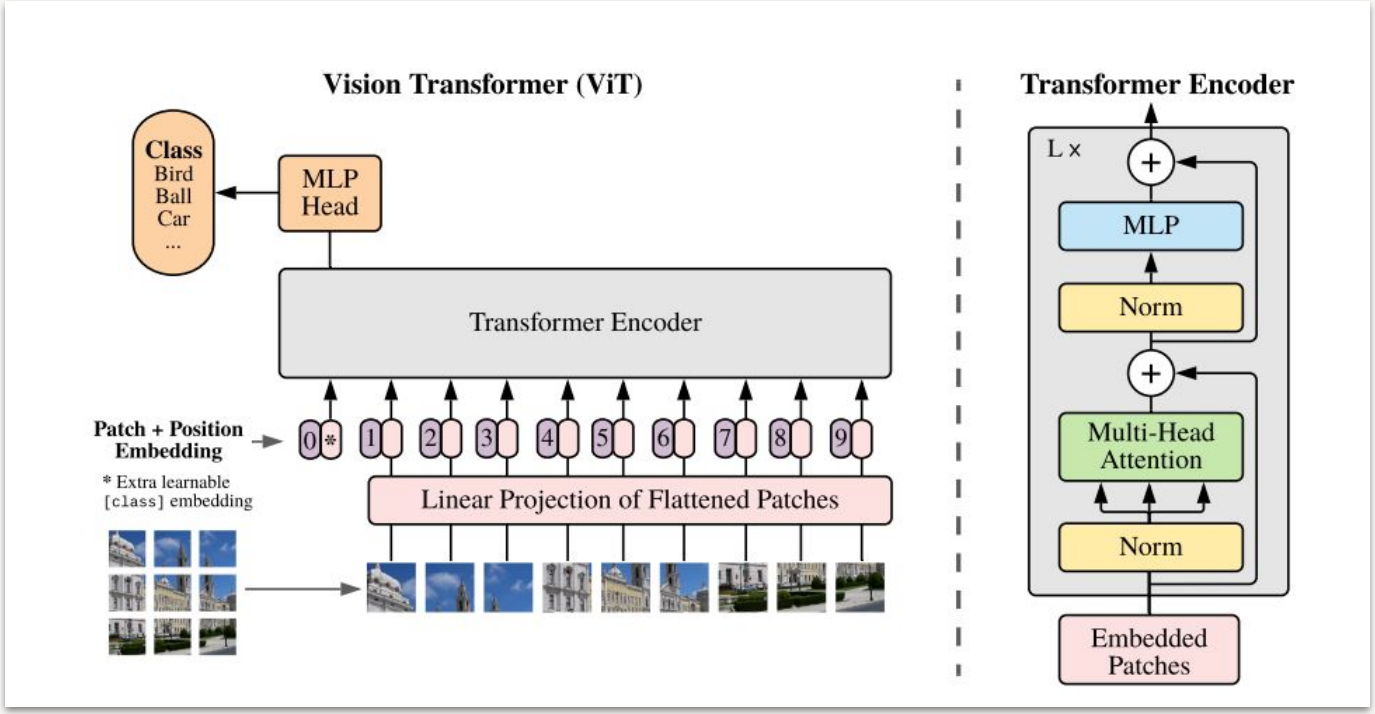
Google Research, Brain Team

{adosovitskiy, neilhoulsby}@google.com

ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.[1]

VISION TRANSFORMER (ViT)



An Image is Worth 16x16 Words

Aplicaciones y casos de uso

MICROSCOPY IMAGES PAPER

https://ieeexplore.ieee.org/document/10178851

Variational Autoencoder as a Data Augmentation tool for Confocal Microscopy Images

1st Eugenia Sol Piñeiro
Dpto. Ingeniería Informática
Instituto Tecnológico de Buenos Aires
Buenos Aires, Argentina
epineiro@itba.edu.ar

2nd Rodrigo Ramele
Dpto. Ingeniería Informática
Instituto Tecnológico de Buenos Aires
Buenos Aires, Argentina
rramele@itba.edu.ar

3rd Juliana Gambini
Centro de Investigación CIDIA - CEPSI
Universidad Nacional de Hurlingham
Universidad Tecnológica Nacional
Universidad Nacional de Tres de Febrero
Buenos Aires, Argentina
juliana.gambini@unahur.edu.ar

***Abstract*—Retinoblastoma is an ocular tumor characterized by malignant cells in the retina of the eye. For its treatment, doctors apply different methods, one of them is the localized injection of a chemotherapy drug called Topotecan. Scientists in the area of medicine are interested in study the speed of Topotecan penetration in tumor clusters. In order to do this, several sequences of microscopy images of Retinoblastoma cell cultures were taken. However, this process is very complex because Retinoblastoma's tumor clusters require stringent growth and maintenance conditions. That is why there is only one reduced set of images, unique in Argentina. Nonetheless, for this same reason statistical studies applied to uncover this penetration dynamics are unreliable.**

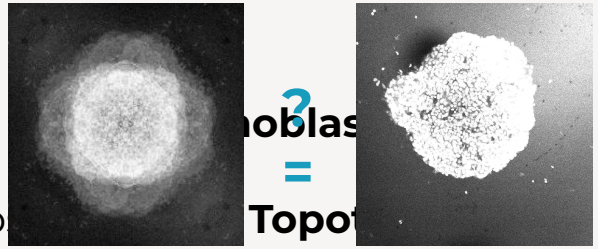
out a predictive analysis and evaluate the results obtained for drawing conclusions.

Hence, in this work we are proposing the usage of Variational Autoencoder to augment this dataset, aiming to imitate the process of Topotecan penetration, while using Transformers to verify that this imitation preserves the dynamics found in the original dataset.

II. Background

Deep Learning is a very successful artificial intelligence

MICROSCOPY IMAGES PAPER



- Imágenes de noblas
- Conservar la visión de los niños afectados Topo
- **Necesidad:** analizar la dinámica de penetración de Topotecan en cultivos de células primarias.
- **Problema:** el cultivo de estas células requiere condiciones muy estrictas

Objetivos:

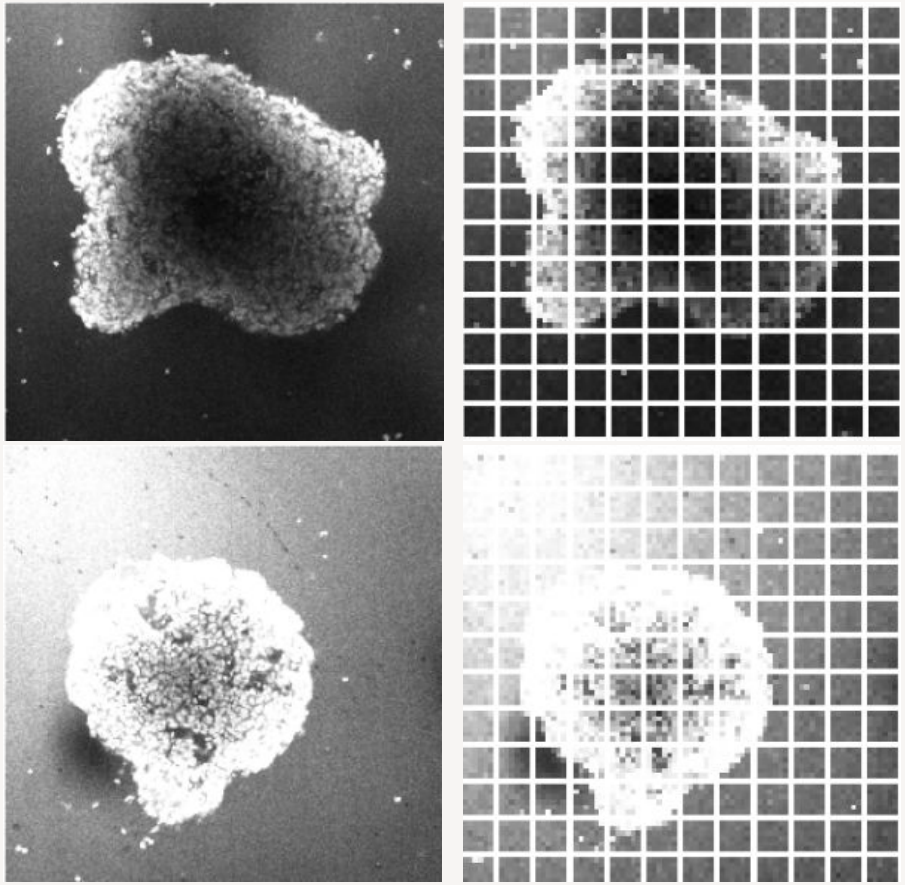
  - **Generar imágenes** sintéticas con las mismas características de las originales porque el dataset es pequeño
  - **Imitar** el proceso de **penetración** del Topotecan.

E. S. Piñeiro, et al, "Variational Autoencoder as a Data Augmentation tool for Confocal Microscopy Images," 2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS), L'Aquila, Italy, 2023, pp. 882-885, doi: 10.1109/CBMS58004.2023.00336.

CLASIFICACIÓN

- Imágenes 128x128
- Patches

Se evaluaron los resultados aplicando técnicas de Computer Vision y clasificación con Transformers



E. S. Piñeiro, R. Ramele and J. Gambini, "Variational Autoencoder as a Data Augmentation tool for Confocal Microscopy Images," 2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS), L'Aquila, Italy, 2023, pp. 882-885, doi: 10.1109/CBMS58004.2023.00336.

OCT IMAGES PAPER

https://ieeexplore.ieee.org/document/11058724

A Transformer-Based Anomaly Detection System for OCT Image Embeddings

1st Eugenia Sol Piñeiro
Dpto. Ingeniería Informática
Instituto Tecnológico de Buenos Aires
Buenos Aires, Argentina
epineiro@itba.edu.ar

2nd Ariel Schlaen
Sección Uveítis, Servicio de Oftalmología
Hospital de Clinicas "José de San Martín"
Universidad de Buenos Aires
Buenos Aires, Argentina
ariel.schlaen@gmail.com

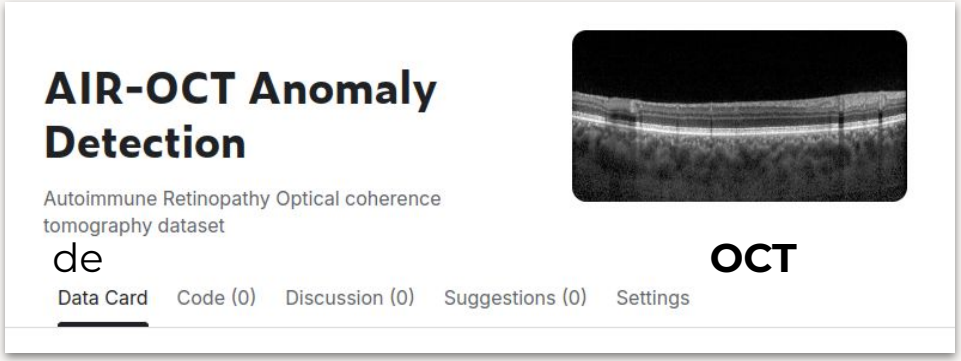
3rd Rodrigo Ramele
Dpto. Ingeniería Informática
Instituto Tecnológico de Buenos Aires
Buenos Aires, Argentina
rramele@itba.edu.ar

4th Juliana Gambini
LIDEC, UNaHur
Universidad Nacional de Hurlingham
Universidad Tecnológica Nacional
Pcia. de Buenos Aires, Argentina
juliana.gambini@unahur.edu.ar

***Abstract*—Retinopathy is characterized by pathological alterations in the retina that can lead to partial or complete vision loss. It can result from a variety of causes, including diabetes, hypertension, and autoimmune disorders, such as Autoimmune Retinopathy, which is particularly challenging to detect and diagnose. Due to its visually similar features, Autoimmune**

visual transduction by converting light into electrical signals. These signals travel through the optic nerve to the brain, where they are interpreted to form visual images. The disruption of the outer retina can result in vision loss, which may become a complete loss if not identified and treated at an early stage.

OCT IMAGES PAPER



- Imágenes

- **Necesidad:** Detectar Retinopatía Autoinmune
- **Problema:** Enfermedad con características similares a otras pero tratamientos muy diferentes → pérdida total de visión

  Objetivos:
  - **Distinguir** entre AIR vs Normal
  - **Dataset** público.

E. S. Piñeiro, et al, "A Transformer-Based Anomaly Detection System for OCT Image Embeddings," 2025 IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS), Madrid, Spain, 2025, pp. 969-972, doi: 10.1109/CBMS65348.2025.00193.

Kaggle AIR Dataset

DALL-E 3

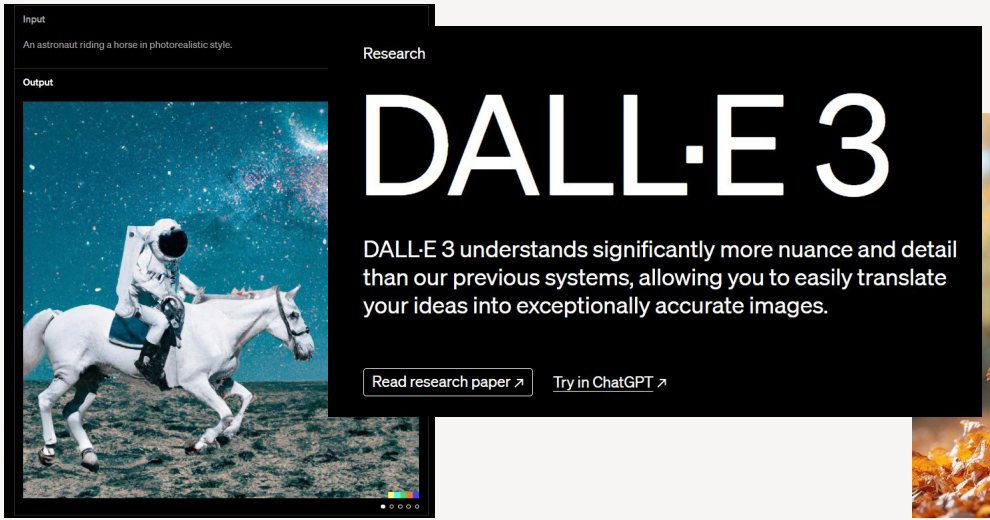
Enero 2021 — DALL-E

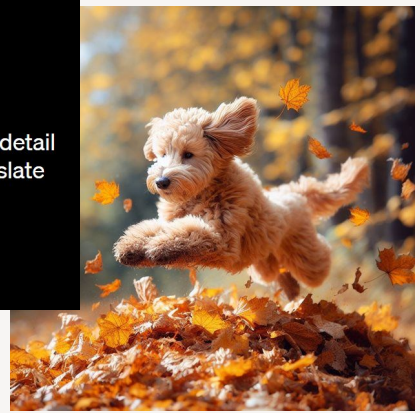
Enero 2022 — **CLIP**

Abril 2022 — DALL-E 2

Agosto 2023 — DALL-E 3

Octubre 2023 — DALL-E 3 + ChatGPT y BingChat





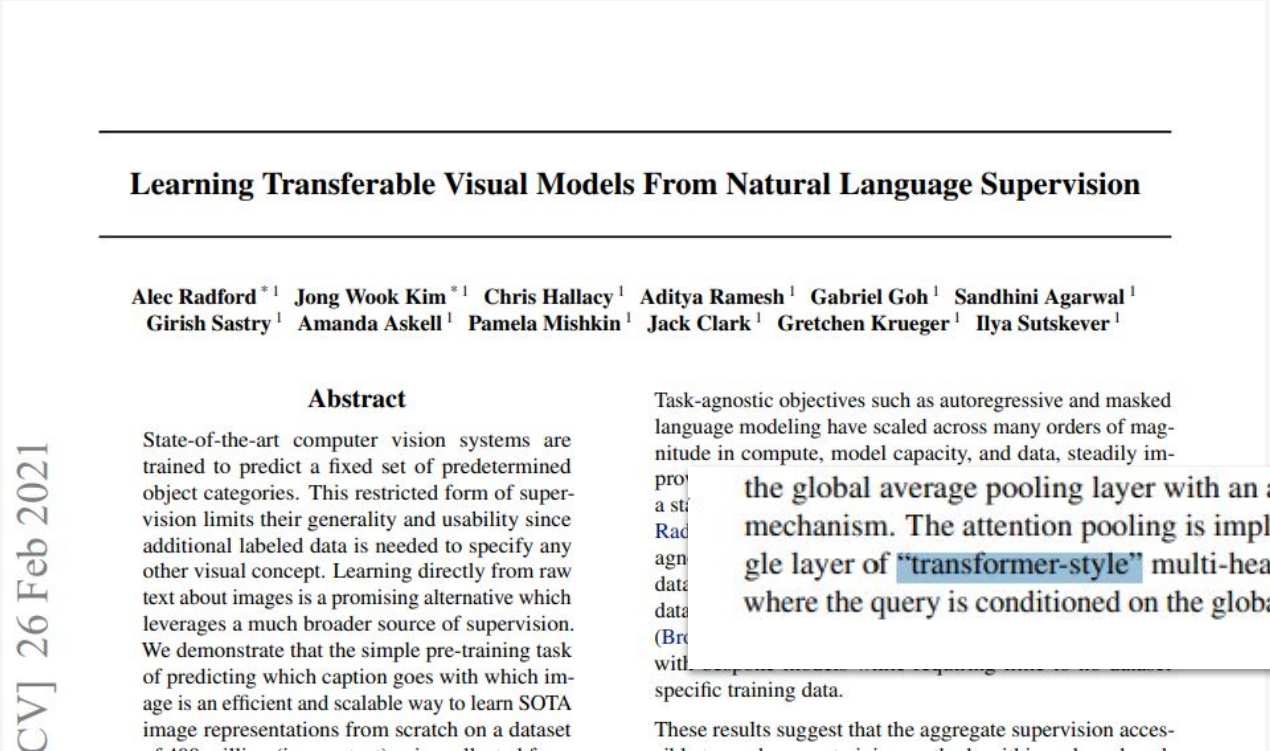
[DALL-E 3]

[Ejemplos]

CLIP

Contrastive Language-Image Pre-Training

https://arxiv.org/pdf/2103.00020



Learning Transferable Visual Models From Natural Language Supervision

Alec Radford [*1] Jong Wook Kim [*1] Chris Hallacy [1] Aditya Ramesh [1] Gabriel Goh [1] Sandhini Agarwal [1]
Girish Sastry [1] Amanda Askell [1] Pamela Mishkin [1] Jack Clark [1] Gretchen Krueger [1] Ilya Sutskever [1]

[cs.CV] 26 Feb 2021

Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset

Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily im-

the global average pooling layer with an attention pooling mechanism. The attention pooling is implemented as a single layer of "transformer-style" multi-head QKV attention where the query is conditioned on the global average-pooled

specific training data.

These results suggest that the aggregate supervision acces-

CLIP

Contrastive Language-Image Pre-Training

Red neuronal entrenada en alrededor de 400M de pares (image-caption)

Función objetivo:

- Maximizar la similitud coseno entre pares image-caption correctos
- Minimizar la similitud coseno entre pares incorrectos

Contrastive Learning

- Las instancias que exhiben similitudes deben estar estrechamente alineadas en el espacio de embeddings aprendido
- Aquellas que son diferentes deben ubicarse más separadas.

CLIP

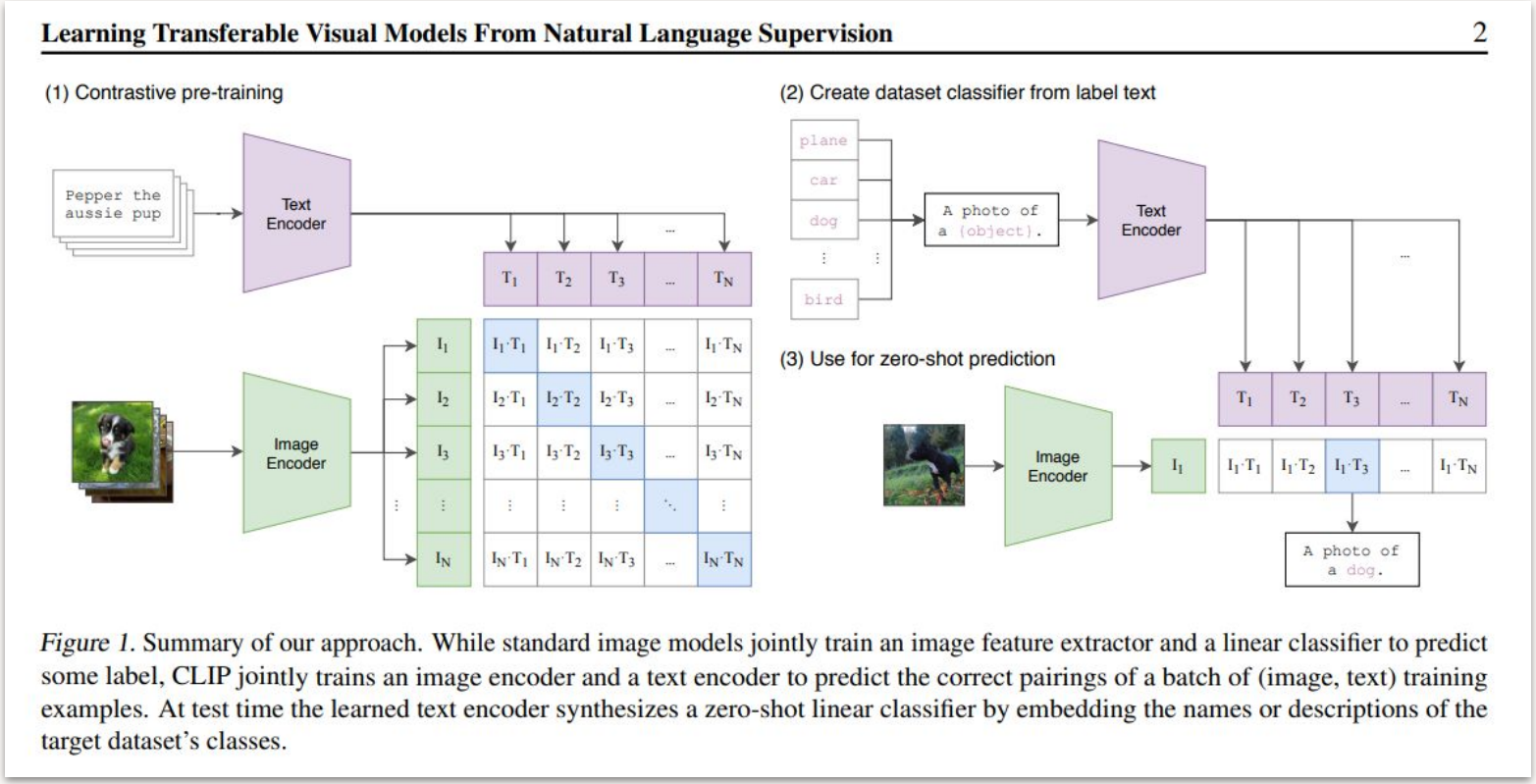
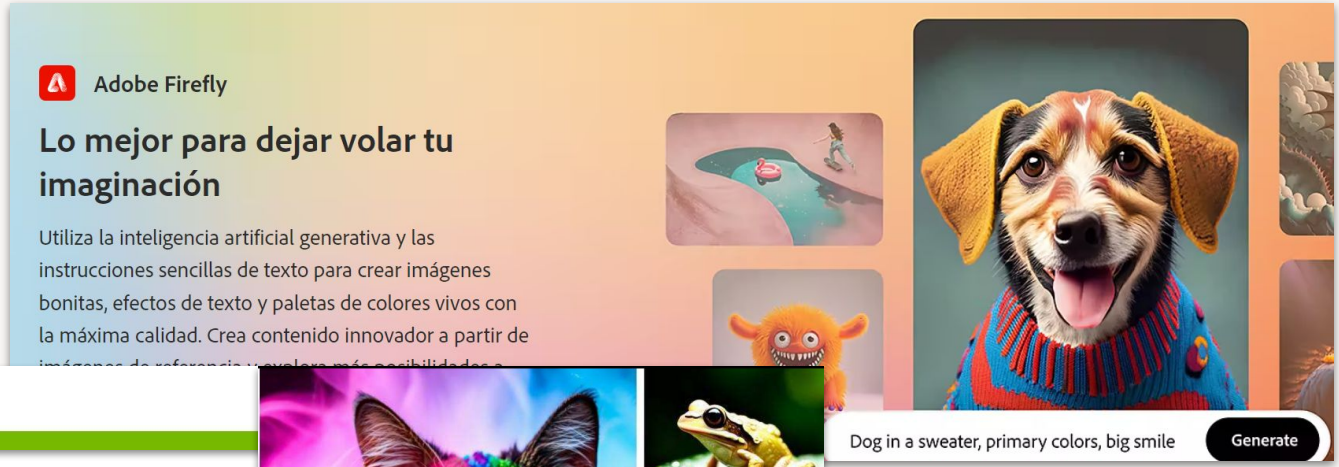
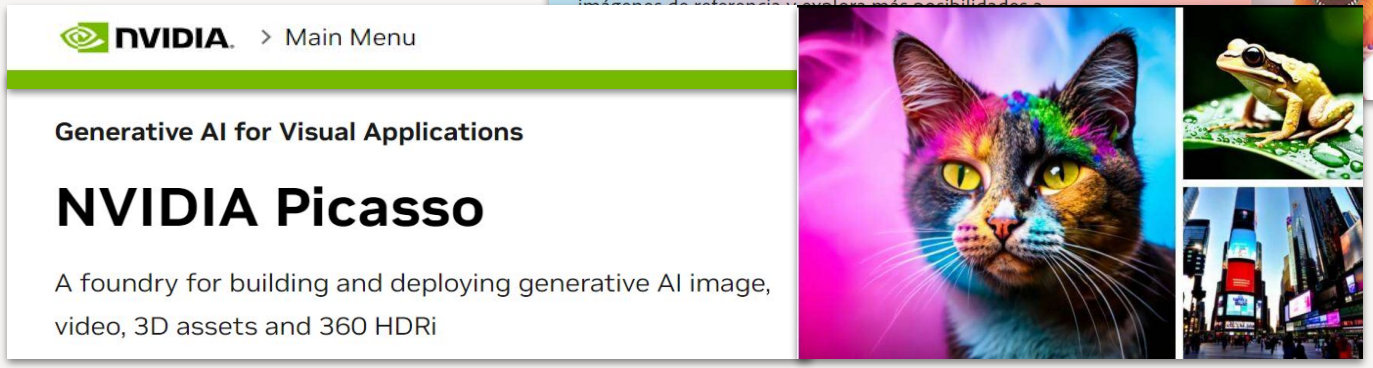


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

IMAGE GENERATION

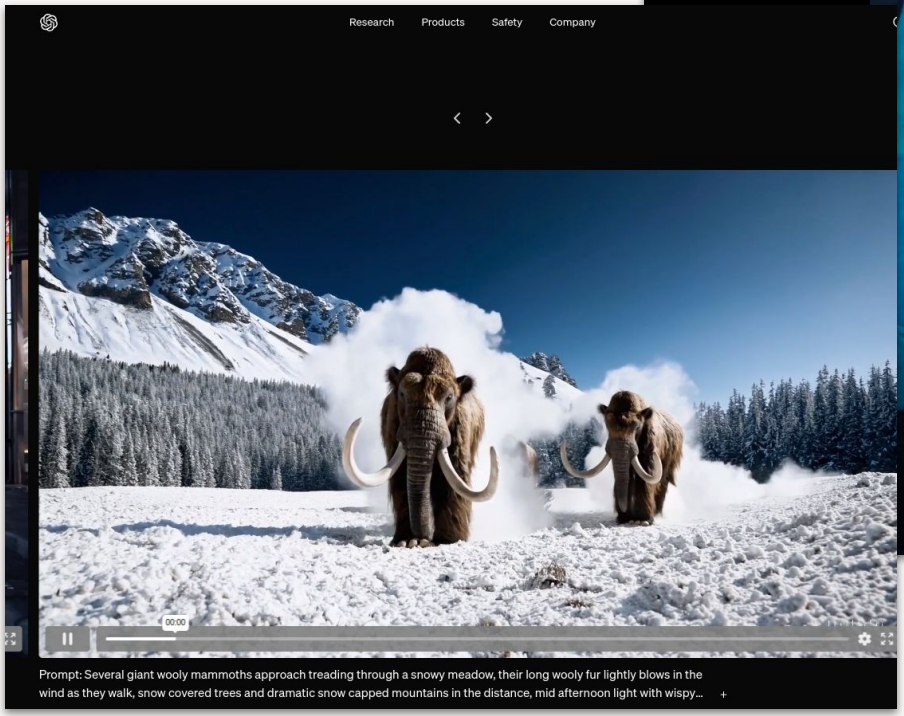


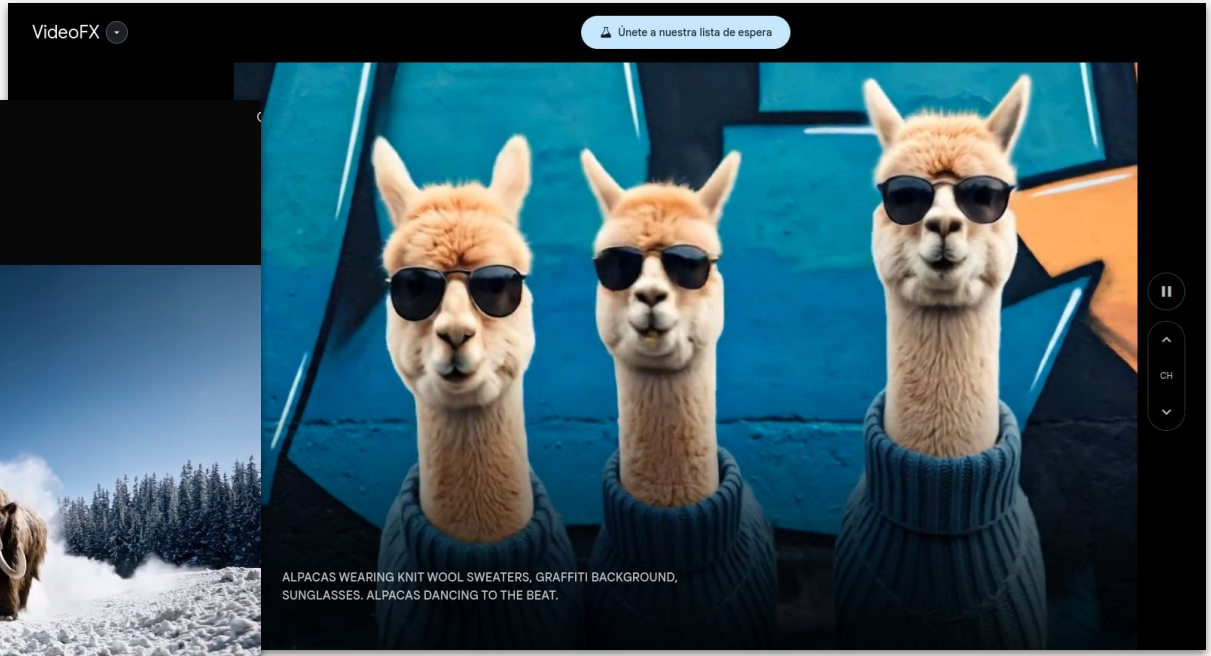


VIDEO GENERATION

[OpenAI - Sora]

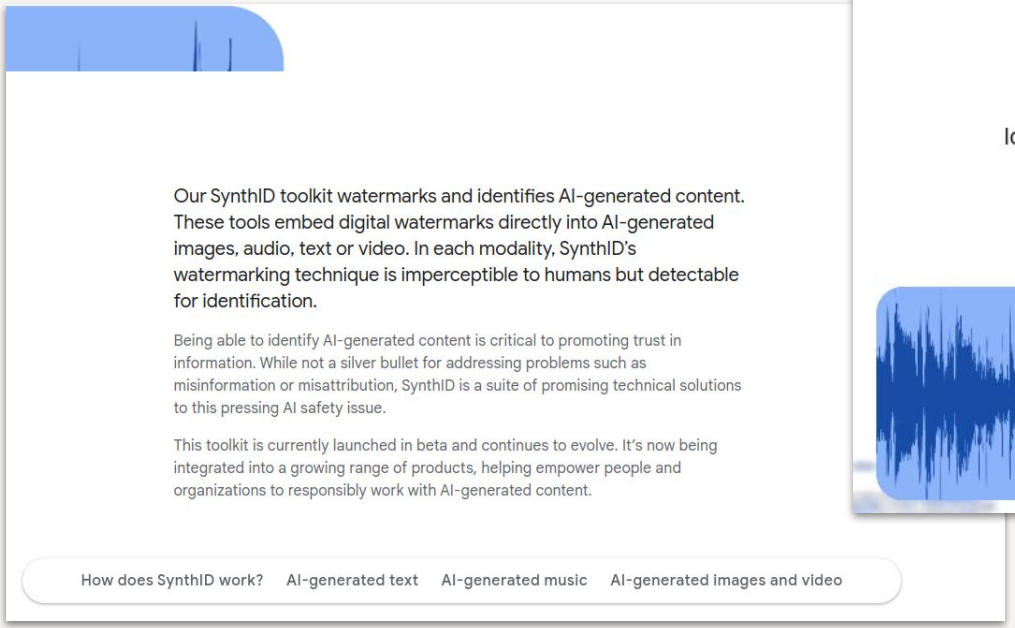
[Google Deepmind - VideoFX]

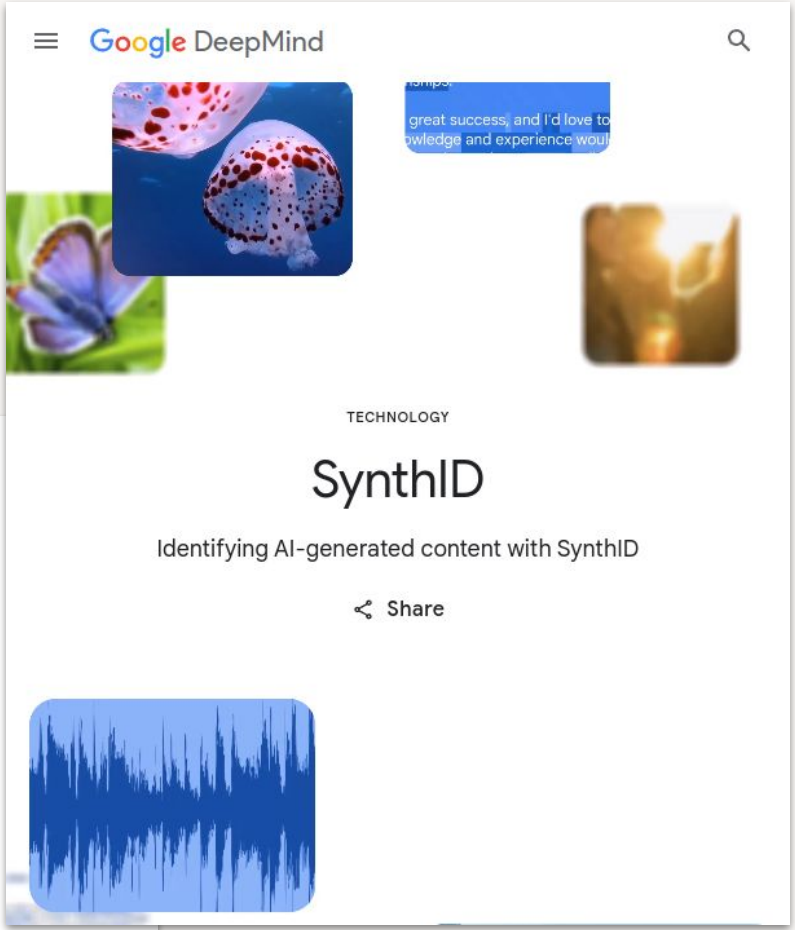




SYNTHID

¿Cómo identificar contenido generado artificialmente?

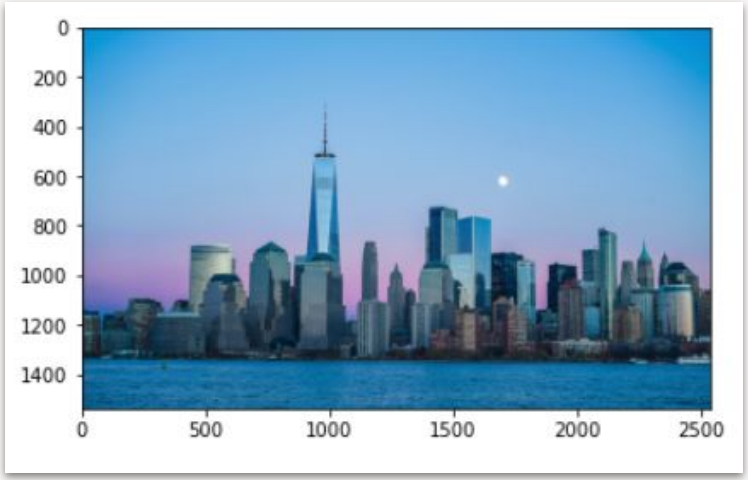




¿Cómo representamos una imagen en embeddings?

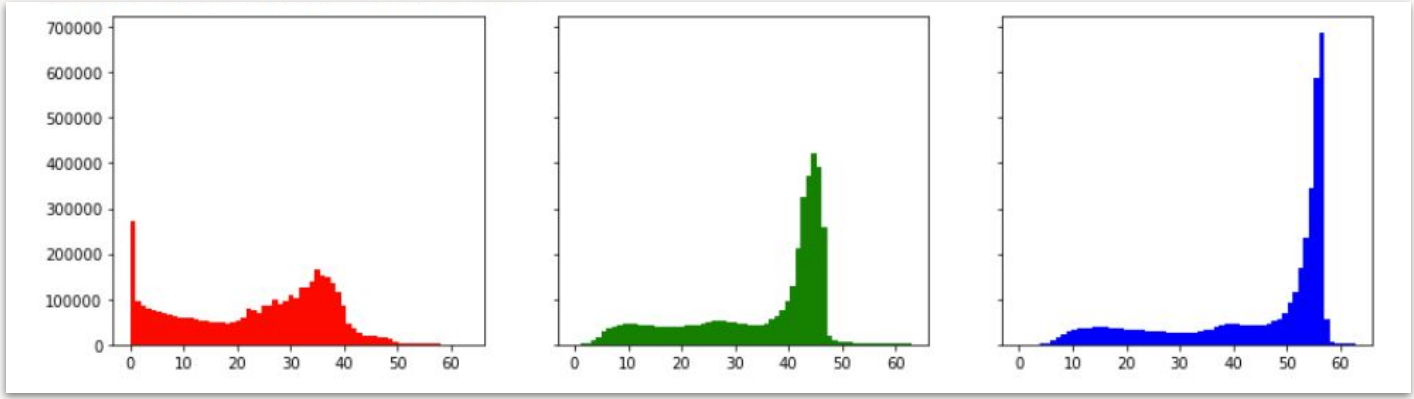
CBIR

Content-**B**ased **I**mage **R**etrieval



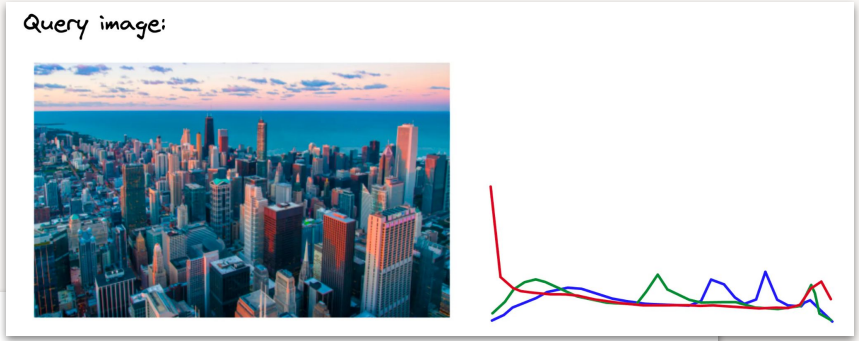
Comparar imágenes en base a su color, objetos, sentido semántico, entre otros.

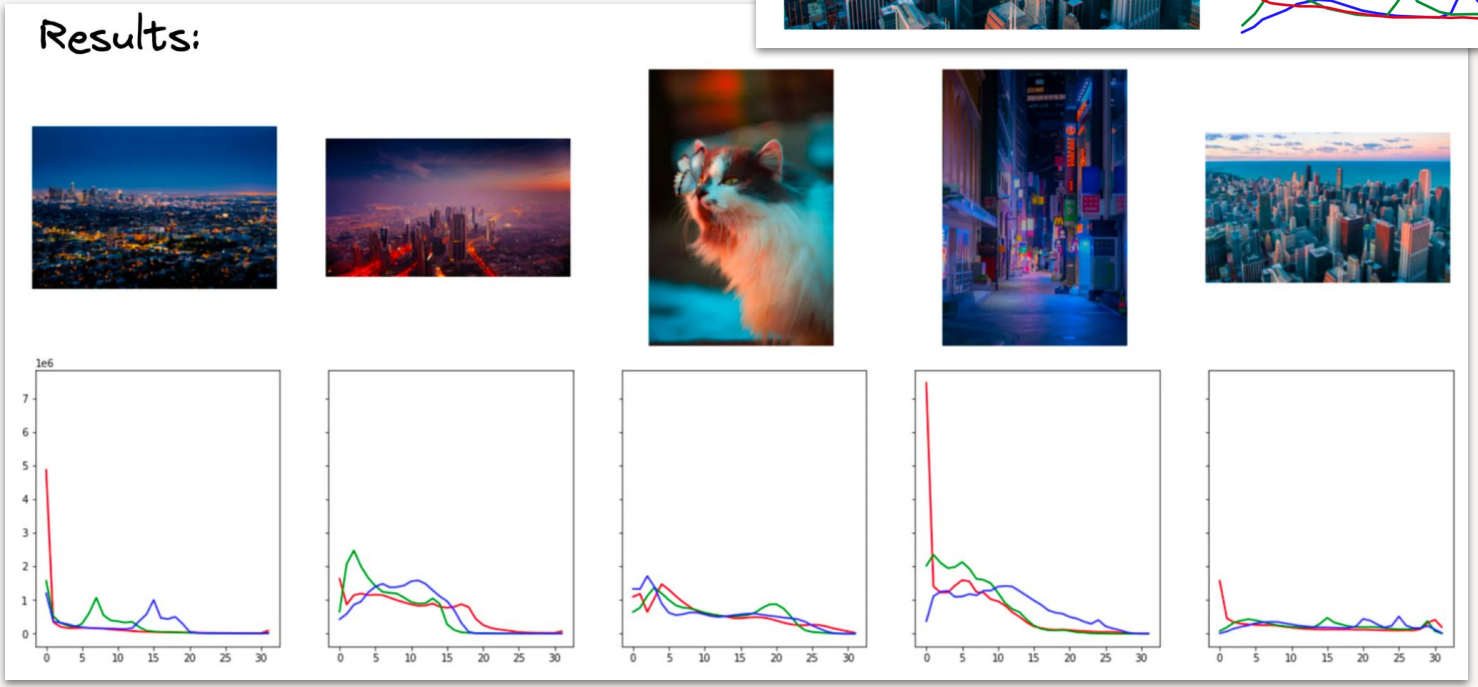
Ejemplo: **Histogramas**



Briggs, J., & Carnevali, L. (n.d.). ***Embedding methods for image search***. Pinecone. Retrieved July 30, 2024

CBIR

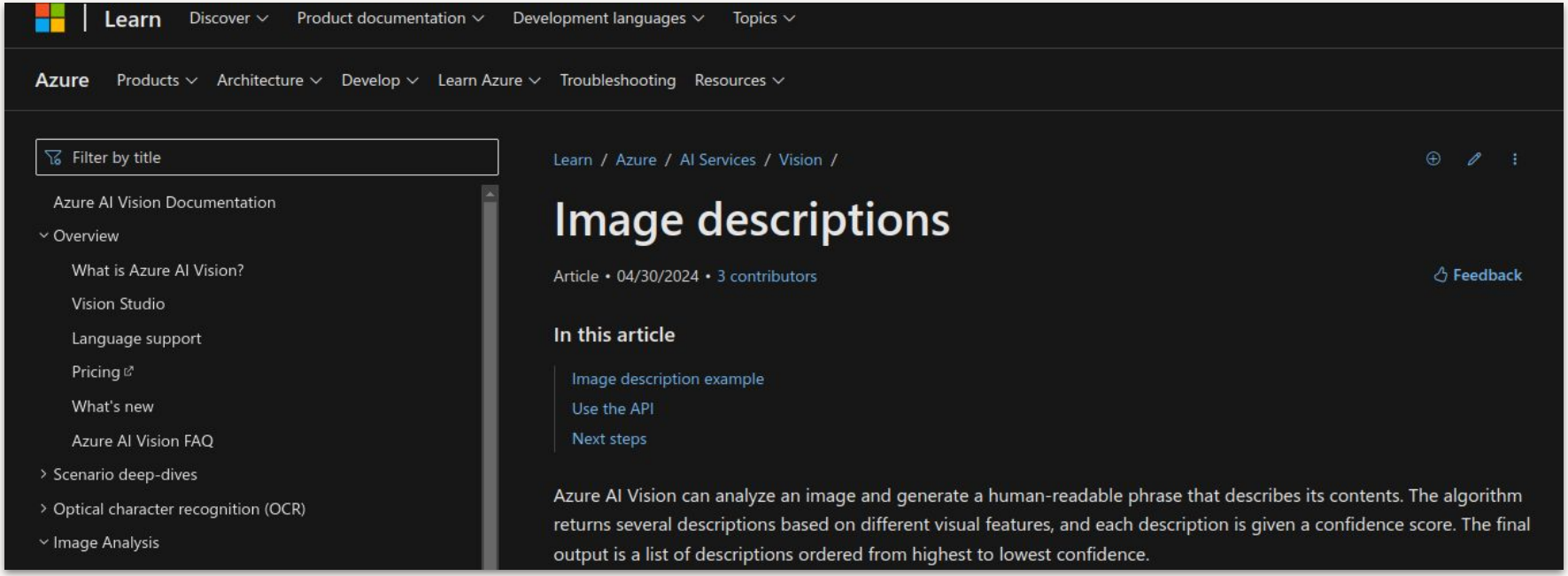




Briggs, J., & Carnevali, L. (n.d.). ***Embedding methods for image search***. Pinecone. Retrieved July 30, 2024

AZURE AI-VISION

Describir con texto una imagen
[Azure Image Descriptions API]



OCR

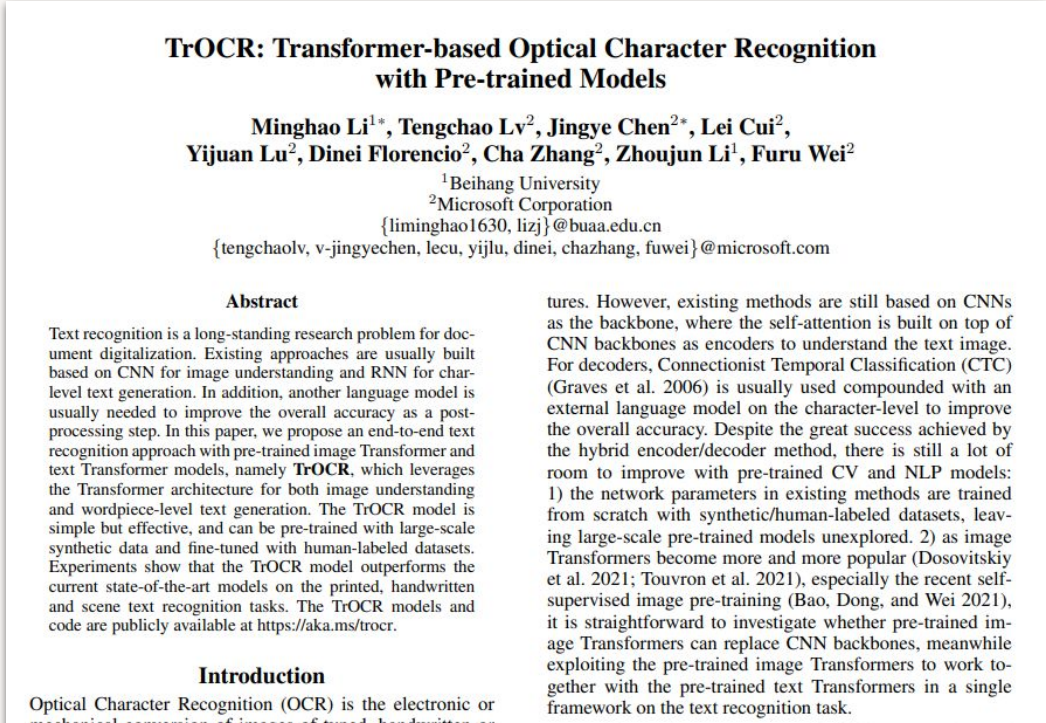
Optical **C**haracter **R**ecognition

Extraer texto de una imagen

[TransformerOCR]

[Keras OCR]

[pytesseract]



TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models

Minghao Li[1*], Tengchao Lv[2], Jingye Chen[2*], Lei Cui[2], Yijuan Lu[2], Dinei Florencio[2], Cha Zhang[2], Zhoujun Li[1], Furu Wei[2]

[1]Beihang University
[2]Microsoft Corporation
{liminghao1630, lizj}@buaa.edu.cn
{tengchaolv, v-jingyechen, lecu, yijlu, dinei, chazhang, fuwei}@microsoft.com

Abstract

Text recognition is a long-standing research problem for document digitalization. Existing approaches are usually built based on CNN for image understanding and RNN for char-level text generation. In addition, another language model is usually needed to improve the overall accuracy as a post-processing step. In this paper, we propose an end-to-end text recognition approach with pre-trained image Transformer and text Transformer models, namely **TrOCR**, which leverages the Transformer architecture for both image understanding and wordpiece-level text generation. The TrOCR model is simple but effective, and can be pre-trained with large-scale synthetic data and fine-tuned with human-labeled datasets. Experiments show that the TrOCR model outperforms the current state-of-the-art models on the printed, handwritten and scene text recognition tasks. The TrOCR models and code are publicly available at https://aka.ms/trocr.

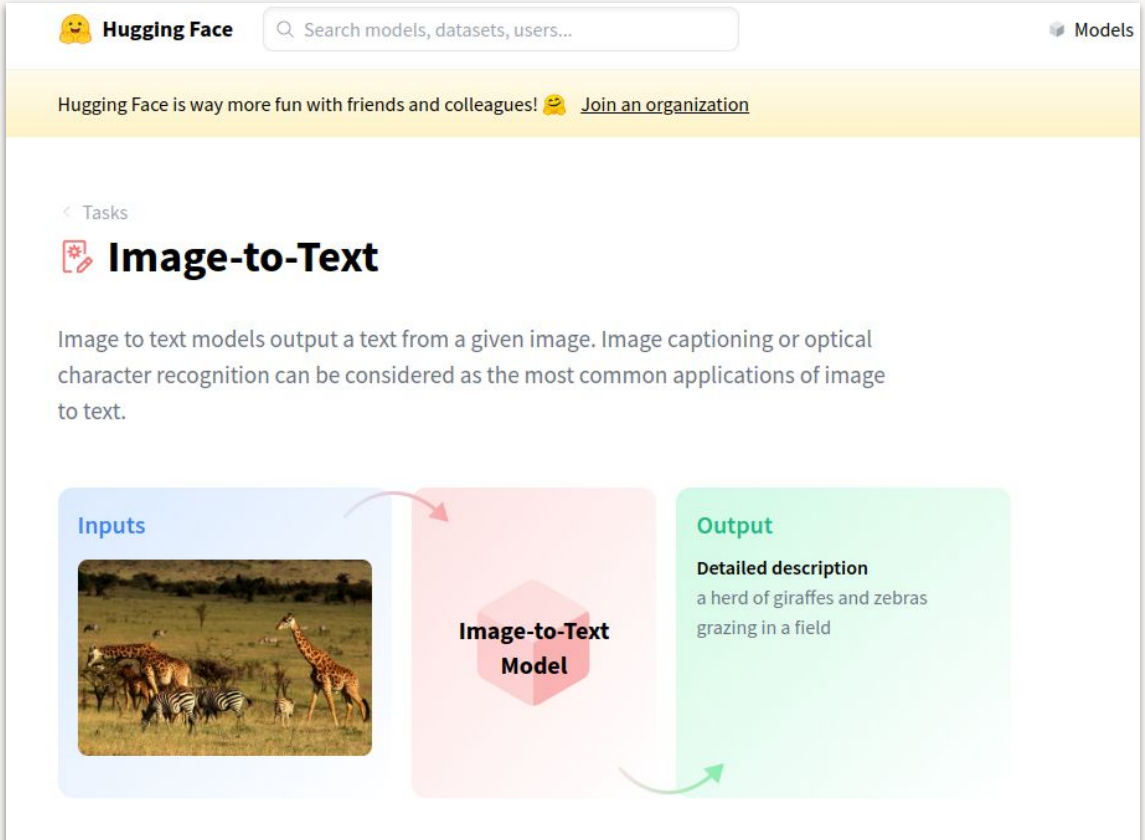
Introduction

Optical Character Recognition (OCR) is the electronic or

tures. However, existing methods are still based on CNNs as the backbone, where the self-attention is built on top of CNN backbones as encoders to understand the text image. For decoders, Connectionist Temporal Classification (CTC) (Graves et al. 2006) is usually used compounded with an external language model on the character-level to improve the overall accuracy. Despite the great success achieved by the hybrid encoder/decoder method, there is still a lot of room to improve with pre-trained CV and NLP models: 1) the network parameters in existing methods are trained from scratch with synthetic/human-labeled datasets, leaving large-scale pre-trained models unexplored. 2) as image Transformers become more and more popular (Dosovitskiy et al. 2021; Touvron et al. 2021), especially the recent self-supervised image pre-training (Bao, Dong, and Wei 2021), it is straightforward to investigate whether pre-trained image Transformers can replace CNN backbones, meanwhile exploiting the pre-trained image Transformers to work together with the pre-trained text Transformers in a single framework on the text recognition task.

HF API

[HF Image to Text]



¿Hay otros métodos?

CLIP
CNN
ViT
SIFT

CASO DE USO - IMAGE RAG

“Product Analyzer”

Se desea construir una aplicación que permite a los usuarios subir **imágenes** de alimentos u otros productos y hacer preguntas sobre ellos.

Objetivo:

- Identificar atributos del producto, como la presencia de lactosa, gluten, azúcar u otros ingredientes.
- Ayudar a tomar decisiones informadas, considerando restricciones alimentarias y condiciones de salud como intolerancia a la **lactosa**, **celiaquía**, **diabetes,** entre otras.

CASO DE USO

¿Cómo genero image embeddings?

```
model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32")
```

Indexing Pipeline

1. Loader
2. Image preprocessing
   (resize, denoise, patches, augmentation, etc)
3. CLIP Embeddings
4. Load to vector db

User input: new image

New image preprocessing and CLIP embeddings
Retrieval and Augmentation

CASO DE USO













Almacenamiento en BD

CASO DE USO

Posibles prompts











02

AUDIO

Aplicaciones

WHISPER

[Open AI - Whisper model]

Robust Speech Recognition via Large-Scale Weak Supervision

Alec Radford [*1] Jong Wook Kim [*1] Tao Xu [1] Greg Brockman [1] Christine McLeavey [1] Ilya Sutskever [1]

Abstract

We study the capabilities of speech processing systems trained simply to predict large amounts of transcripts of audio on the internet. When scaled to 680,000 hours of multilingual and multitask supervision, the resulting models generalize well to standard benchmarks and are often competitive with prior fully supervised results but in a zero-shot transfer setting without the need for any fine-tuning. When compared to humans, the models approach their accuracy and robustness. We are releasing models and inference code to serve as a foundation for further work on robust speech processing.

1. Introduction

Progress in speech recognition has been energized by the development of unsupervised pre-training techniques exemplified by Wav2Vec 2.0 (Baevski et al., 2020). Since these methods are exceedingly adept at finding patterns within a training dataset which boost performance on held-out data from the same dataset. However, some of these patterns are brittle and spurious and don't generalize to other datasets and distributions. In a particularly disturbing example, Radford et al. (2021) documented a 9.2% increase in object classification accuracy when fine-tuning a computer vision model on the ImageNet dataset (Russakovsky et al., 2015) without observing any improvement in average accuracy when classifying the same objects on seven other natural image datasets. A model that achieves "superhuman" performance when trained on a dataset can still make many basic errors when evaluated on another, possibly precisely because it is exploiting those dataset-specific quirks that humans are oblivious to (Geirhos et al., 2020).

This suggests that while unsupervised pre-training has improved the quality of audio encoders dramatically, the lack of an equivalently high-quality pre-trained decoder, combined with a recommended protocol of dataset-specific fine-tuning, is a crucial weakness which limits their usefulness and robustness. The goal of a speech recognition system

GENERACIÓN DE MÚSICA

Music Transformer: Generating Music with Long-Term Structure

MUSIC TRANSFORMER: GENERATING MUSIC WITH LONG-TERM STRUCTURE

Cheng-Zhi Anna Huang* **Ashish Vaswani** **Jakob Uszkoreit** **Noam Shazeer**
Ian Simon **Curtis Hawthorne** **Andrew M. Dai** **Matthew D. Hoffman**
Monica Dinculescu **Douglas Eck**
Google Brain

ABSTRACT

Music relies heavily on repetition to build structure and meaning. Self-reference occurs on multiple timescales, from motifs to phrases to reusing of entire sections of music, such as in pieces with ABA structure. The Transformer (Vaswani et al., 2017), a sequence model based on self-attention, has achieved compelling results in many generation tasks that require maintaining long-range coherence. This suggests that self-attention might also be well-suited to modeling music. In musical composition and performance, however, relative timing is critically important. Existing approaches for representing relative positional information in the Transformer modulate attention based on pairwise distance (Shaw et al., 2018). This is impractical for long sequences such as musical compositions since their memory complexity for intermediate relative information is quadratic in the sequence length. We propose an algorithm that reduces their intermediate memory requirement to linear in the sequence length. This enables us to demonstrate that a Transformer with our modified relative attention mechanism can generate minute-long compositions (thousands of steps, four times the length modeled in Oore et al. (2018)) with compelling structure, generate continuations that coherently elaborate

SPEECH T5

SpeechT5: automatic speech recognition, speech synthesis, speech translation, voice conversion, speech enhancement, and speaker identification

SpeechT5: Unified-Modal Encoder-Decoder Pre-Training for Spoken Language Processing

Junyi Ao[1,2,*], Rui Wang[3,*], Long Zhou[4,*], Chengyi Wang[4], Shuo Ren[4], Yu Wu[4], Shujie Liu[4], Tom Ko[1], Qing Li[2], Yu Zhang[1,5], Zhihua Wei[3], Yao Qian[4], Jinyu Li[4], Furu Wei[4]

[1]Department of Computer Science and Engineering, Southern University of Science and Technology
[2]Department of Computing, The Hong Kong Polytechnic University
[3]Department of Computer Science and Technology, Tongji University
[4]Microsoft [5]Peng Cheng Laboratory

Abstract

Motivated by the success of T5 (Text-To-Text Transfer Transformer) in pre-trained natural language processing models, we propose a unified-modal SpeechT5 framework that explores the encoder-decoder pre-training for self-supervised speech/text representation learning. The SpeechT5 framework consists of a shared encoder-decoder network and six modal-specific (speech/text) pre/post-nets. After preprocessing the input speech/text through

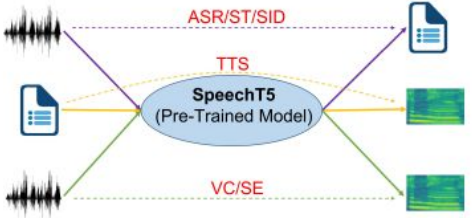


Figure 1: An illustration of the SpeechT5 framework, which treats spoken language processing tasks as a speech/text to speech/text format, including au-

¿Cómo representamos un audio en embeddings?

ASR

Automatic Speech Recognition

Interpretar y transcribir el lenguaje hablado en texto



OPEN AI TRANSCRIPTIONS API

Transcriptions

The transcriptions API takes as input the audio file you want to transcribe and the desired output file format for the transcription of the audio. We currently support multiple input and output file formats.

Transcribe audio — curl

```
curl --request POST \
  --url https://api.openai.com/v1/audio/transcriptions \
  --header "Authorization: Bearer $OPENAI_API_KEY" \
  --header 'Content-Type: multipart/form-data' \
  --form file=@/path/to/file/audio.mp3 \
  --form model=whisper-1
```

By default, the response type will be json with the raw text included.

```
{
  "text": "Imagine the wildest idea that you've ever had, and you're curious about how it might scale to something that's a 100, a 1,000 times bigger.
....
}
```

OPEN AI TRANSLATIONS API

Translations

The translations API takes as input the audio file in any of the supported languages and transcribes, if necessary, the audio into English. This differs from our /Transcriptions endpoint since the output is not in the original input language and is instead translated to English text.

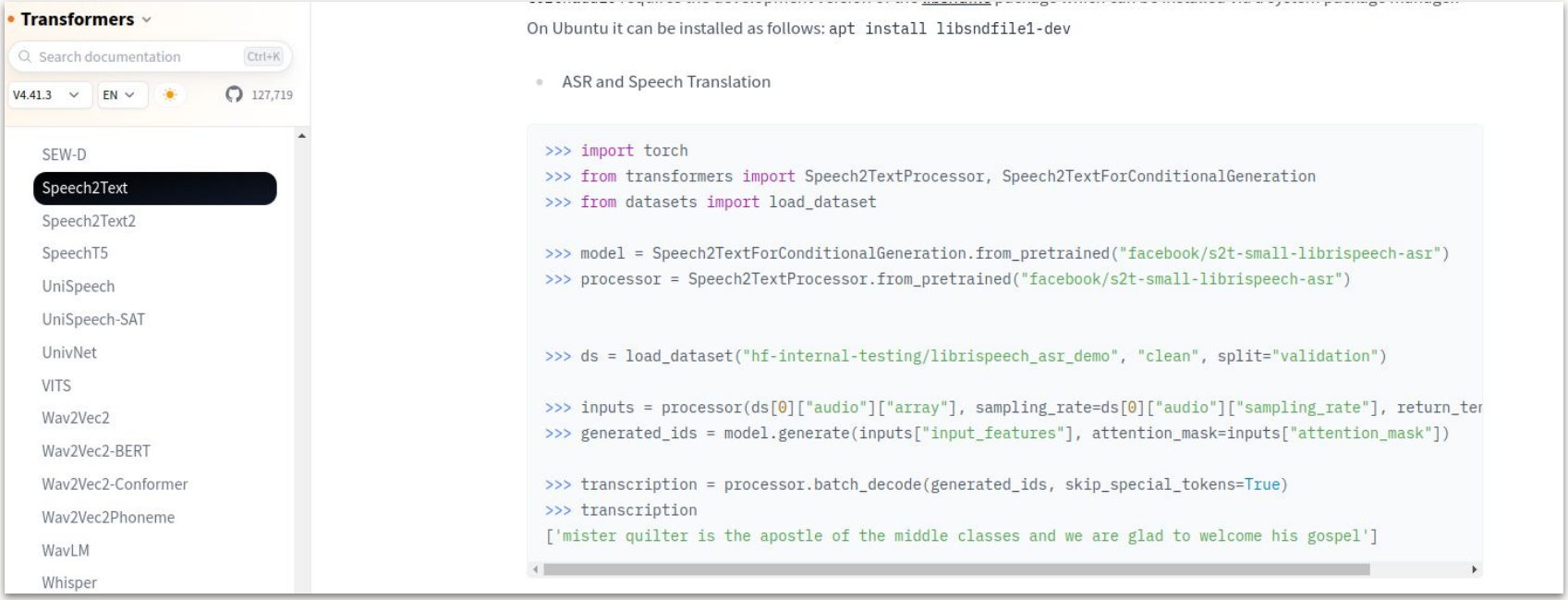
Translate audio — curl

```
curl --request POST \
  --url https://api.openai.com/v1/audio/translations \
  --header "Authorization: Bearer $OPENAI_API_KEY" \
  --header 'Content-Type: multipart/form-data' \
  --form file=@/path/to/file/german.mp3 \
  --form model=whisper-1
```

In this case, the inputted audio was german and the outputted text looks like:

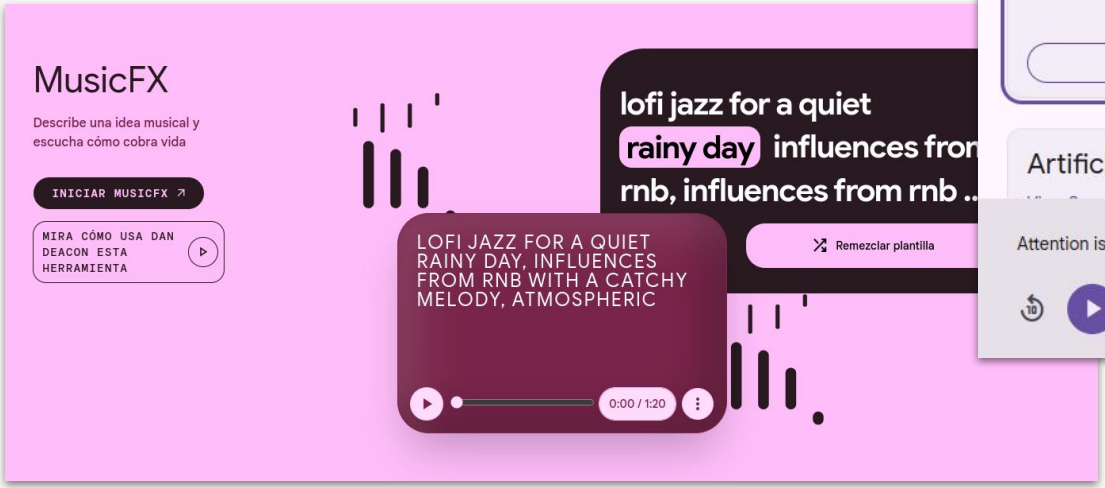
```
Hello, my name is Wolfgang and I come from Germany. Where are you heading today?
```

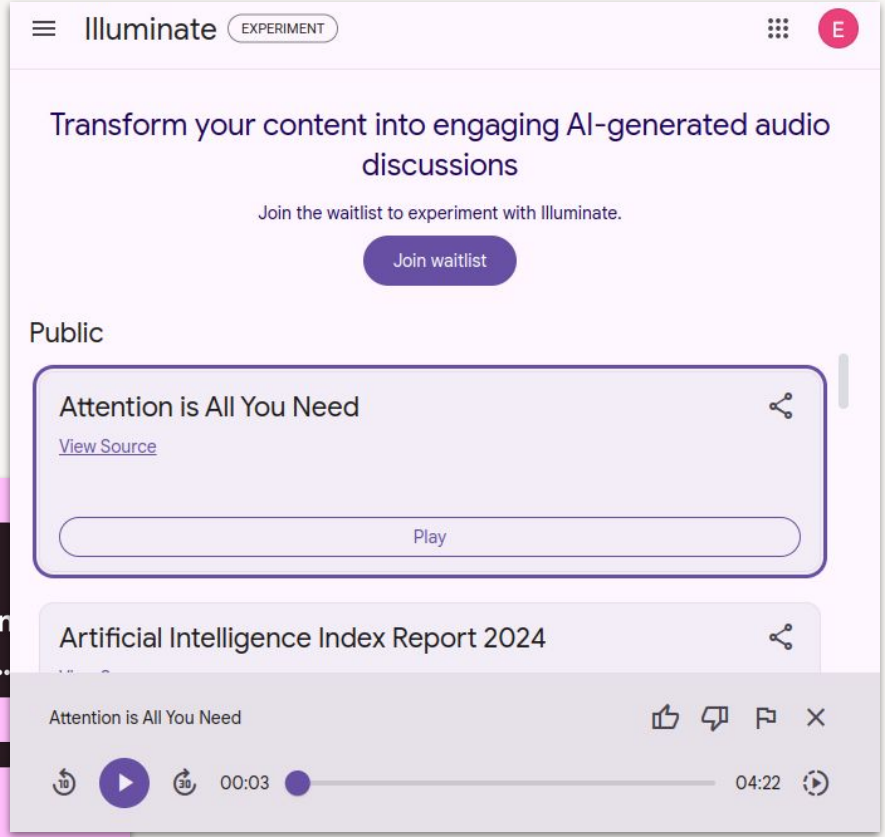
HUGGING FACE STT



AUDIO GENERATION

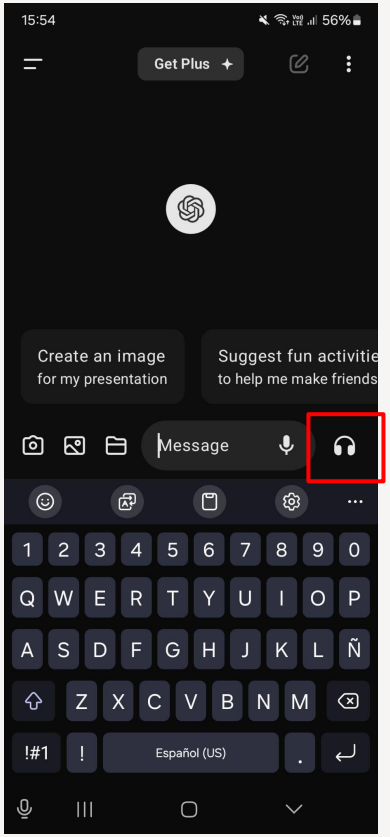
[Google Deepmind - MusicFX]

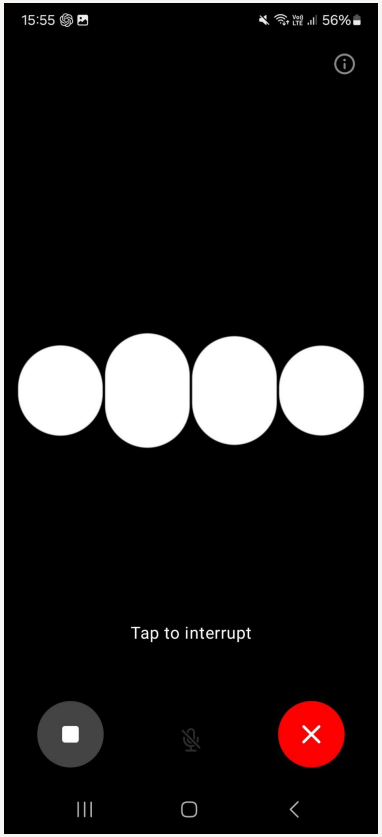




[Google Illuminate]

AUDIO GENERATION





03

MULTIMODAL

Aplicaciones

GPT-4V

[GPT-4v Technical Report]

[GPT-4v Safety Properties]

GPT-4 Technical Report

OpenAI*

Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

1 Introduction

This technical report presents GPT-4, a large multimodal model capable of processing image and text inputs and producing text outputs. Such models are an important area of study as they have the potential to be used in a wide range of applications, such as dialogue systems, text summarization, and machine translation. As such, they have been the subject of substantial interest and progress in recent years [1–34].

One of the main goals of developing such models is to improve their ability to understand and generate natural language text, particularly in more complex and nuanced scenarios. To test its capabilities in such scenarios, GPT-4 was evaluated on a variety of exams originally designed for humans. In

GPT-4V(ision) System Card

OpenAI

September 25, 2023

1 Introduction

GPT-4 with vision (GPT-4V) enables users to instruct GPT-4 to analyze image inputs provided by the user, and is the latest capability we are making broadly available. Incorporating additional modalities (such as image inputs) into large language models (LLMs) is viewed by some as a key frontier in artificial intelligence research and development [1, 2, 3]. Multimodal LLMs offer the possibility of expanding the impact of language-only systems with novel interfaces and capabilities, enabling them to solve new tasks and provide novel experiences for their users.

In this system card, [4, 5] [1] we analyze the safety properties of GPT-4V. Our work on safety for GPT-4V builds on the work done for GPT-4 [7] and here we dive deeper into the evaluations, preparation, and mitigation work done specifically for image inputs.

Similar to GPT-4, training of GPT-4V was completed in 2022 and we began providing early access to the system in March 2023. As GPT-4 is the technology behind the visual capabilities of GPT-4V, its training process was the same. The pre-trained model was first trained to predict the next word in a document, using a large dataset of text and image data from the Internet as well as licensed sources of data. It was then fine-tuned with additional data, using an algorithm called reinforcement learning from human feedback (RLHF),[8, 9] to produce outputs that are preferred by human trainers.

Large multimodal models introduce different limitations and expand the risk surface compared to text-based language models. GPT-4V possesses the limitations and capabilities of each modality (text and vision), while at the same time presenting novel capabilities emerging from the intersection of said modalities and from the intelligence and reasoning afforded by large scale models.

This system card outlines how OpenAI prepared the vision capabilities of GPT-4 for deployment. It describes the early access period of the model for small scale users and safety learnings OpenAI gained from this period, multimodal evaluations built to study the model's fitness for deployment, key findings of expert red teamers, and the mitigations OpenAI implemented prior to broad release.

Llama

Llama 3 Paper

Meta

The Llama 3 Herd of Models

Llama Team, AI @ Meta[1]
[1]A detailed contributor list can be found in the appendix of this paper.

Modern artificial intelligence (AI) systems are powered by foundation models. This paper presents a new set of foundation models, called Llama 3. It is a herd of language models that natively support multilinguality, coding, reasoning, and tool usage. Our largest model is a dense Transformer with 405B parameters and a context window of up to 128K tokens. This paper presents an extensive empirical evaluation of Llama 3. We find that Llama 3 delivers comparable quality to leading language models such as GPT-4 on a plethora of tasks. We publicly release Llama 3, including pre-trained and post-trained versions of the 405B parameter language model and our Llama Guard 3 model for input and output safety. The paper also presents the results of experiments in which we integrate image, video, and speech capabilities into Llama 3 via a compositional approach. We observe this approach performs competitively with the state-of-the-art on image, video, and speech recognition tasks. The resulting models are not yet being broadly released as they are still under development.

Date: July 23, 2024
Website: https://llama.meta.com/

ETAPAS

Llama 3 Paper

- **Managing complexity.** We make design choices that seek to maximize our ability to scale the model development process. For example, we opt for a standard dense Transformer model architecture (Vaswani et al., 2017) with minor adaptations, rather than for a mixture-of-experts model (Shazeer et al., 2017) to maximize training stability. Similarly, we adopt a relatively simple post-training procedure based on supervised finetuning (SFT), rejection sampling (RS), and direct preference optimization (DPO; Rafailov et al. (2023)) as opposed to more complex reinforcement learning algorithms (Ouyang et al., 2022; Schulman et al., 2017) that tend to be less stable and harder to scale.

Pre-training

Multilingual corpus, next token prediction

Post-training

Instruction tuning, Direct Preference Optimization (DPO) , Tool Use, Safety Mitigation

MULTI-MODAL

Llama 3 Paper

Se agrega capacidad de procesar **imágenes** y **audio** por medio de módulos independientes.

Multi-modal encoder pre-training

- **Image Encoder** → entrena en pares (img, txt), relación contenido-descripción
- **Speech Encoder** → self-supervised (masking de partes del speech para luego reconstruirlas)

Vision Adapter training

Integra el image encoder al pre-trained model a través y agrega video-adapter.

Speech Adapter training

Convierte speech a token y lo agregan al modelo

CAPABILITIES / EXPERTS

Llama 3 Paper

CODE
Code generation, documentation, debugging for Python, Java, Javascript, C/C++, Typescript, Rust, PHP, HTML/CSS, SQL, bash/shell.

MULTILINGUALITY
Training an expert specialized on multilingual data

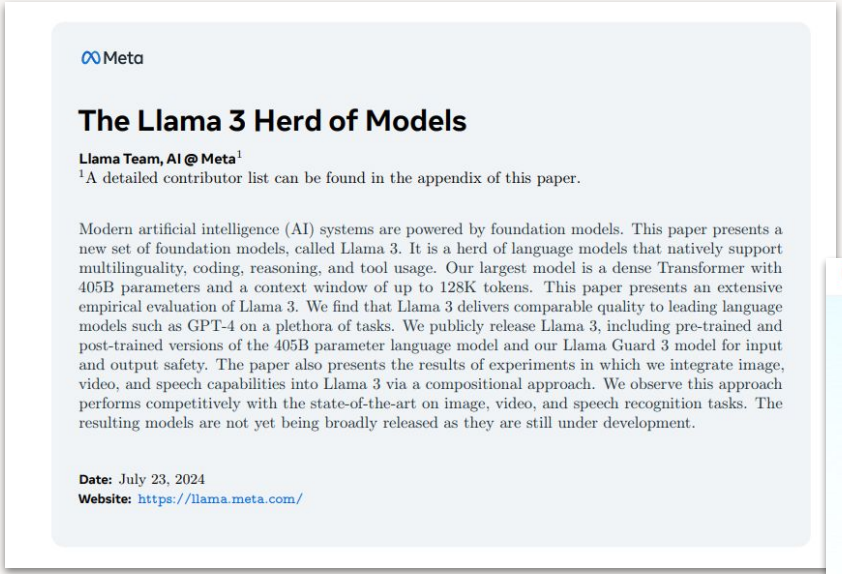
MATH
Multi-step computations and arrive at the correct final answer

TOOL USE
Search Engine, Python Interpreter (ej: read files), Wolfram Alpha API

04

EXTRAS

Video y Paper sugeridos



Meta

The Llama 3 Herd of Models

Llama Team, AI @ Meta[1]
[1]A detailed contributor list can be found in the appendix of this paper.

Modern artificial intelligence (AI) systems are powered by foundation models. This paper presents a new set of foundation models, called Llama 3. It is a herd of language models that natively support multilinguality, coding, reasoning, and tool usage. Our largest model is a dense Transformer with 405B parameters and a context window of up to 128K tokens. This paper presents an extensive empirical evaluation of Llama 3. We find that Llama 3 delivers comparable quality to leading language models such as GPT-4 on a plethora of tasks. We publicly release Llama 3, including pre-trained and post-trained versions of the 405B parameter language model and our Llama Guard 3 model for input and output safety. The paper also presents the results of experiments in which we integrate image, video, and speech capabilities into Llama 3 via a compositional approach. We observe this approach performs competitively with the state-of-the-art on image, video, and speech recognition tasks. The resulting models are not yet being broadly released as they are still under development.

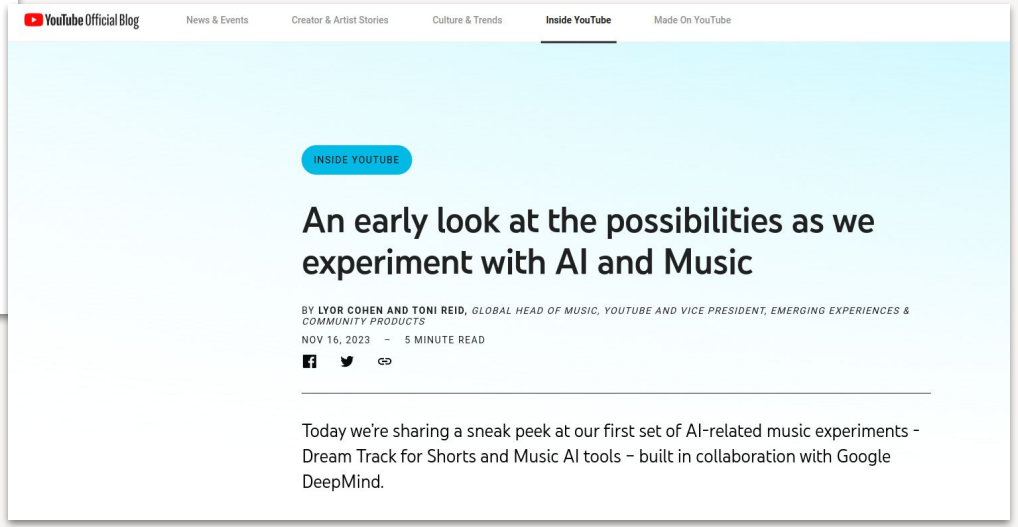
Date: July 23, 2024
Website: https://llama.meta.com/

[Inside Youtube - AI and Music Experiments]

[Lyria - Google DeepMind]

Pregunta abierta

¿Cómo evalúo la performance de mi modelo?



YouTube Official Blog | News & Events | Creator & Artist Stories | Culture & Trends | Inside YouTube | Made On YouTube

INSIDE YOUTUBE

An early look at the possibilities as we experiment with AI and Music

BY LYOR COHEN AND TONI REID, *GLOBAL HEAD OF MUSIC, YOUTUBE AND VICE PRESIDENT, EMERGING EXPERIENCES & COMMUNITY PRODUCTS*
NOV 16, 2023 – 5 MINUTE READ

Today we're sharing a sneak peek at our first set of AI-related music experiments - Dream Track for Shorts and Music AI tools – built in collaboration with Google DeepMind.

05

BIBLIOGRAFIA

BIBLIOGRAFIA

Briggs, J., & Carnevali, L. (n.d.). ***Embedding methods for image search***. Pinecone. Retrieved July 30, 2024, from https://www.pinecone.io/learn/series/image-search/

Keita, Z. (n.d.). ***Text-to-image and image-to-image search using CLIP***. Pinecone. Retrieved July 30, 2024 from https://www.pinecone.io/learn/clip-image-search/

A. Radford, J. W. Kim, et al., ***Learning Transferable Visual Models From Natural Language Supervision (2021)*** https://arxiv.org/pdf/2103.00020v1

Llama Team. 2024, July 23. ***The Llama 3 herd of models***. AI @ Meta. https://llama.meta.com/

OpenAI ***GTP-4 Technical Report***, *2024* https://arxiv.org/pdf/2303.08774